

# DigiChrom ML Pipeline

Vorhersage der galvanischen Chromschichtdicke

Erstellt: 29.04.2026 07:53

## Datensatz

|                 |     |                   |     |
|-----------------|-----|-------------------|-----|
| Gesamt          | 441 | Training          | 374 |
| Testdaten       | 67  | Feature           | 20  |
| Zielwertbereich |     | [0.044, 1.368] µm |     |

### Bester Regressionsmodell

ensemble:weighted

|     |           |      |           |
|-----|-----------|------|-----------|
| R²  | 0.7548    | RMSE | 0.1262 µm |
| MAE | 0.0755 µm |      |           |

### Bester Klassifikator

ensemble:averaging

|               |           |          |        |
|---------------|-----------|----------|--------|
| Schwellenwert | 0.2654 µm | F1-Score | 0.9014 |
| AUC           | 0.9730    | Accuracy | 0.8955 |

## Regression — Schichtdickenvorhersage (µm)

Das Regressions-Modell sagt die Dicke der Chromschicht in µm vorher. Die Pipeline durchläuft Cross-Validation, Hyperparameter-Optimierung (HPO), Ensembling und wählt automatisch den besten Gesamtprädiktor aus.

### 1 • Cross-Validation — Test aller Basismodelle

| Modell                 | R <sup>2</sup><br>Mittel | R <sup>2</sup> Std | RMSE<br>Mittel | RMSE<br>Std | MAE<br>Mittel | MAE<br>Std |
|------------------------|--------------------------|--------------------|----------------|-------------|---------------|------------|
| catboost               | 0.72                     | 0.0583             | 0.0959         | 0.0108      | 0.0586        | 0.0051     |
| extra_trees            | 0.6843                   | 0.0874             | 0.1014         | 0.0149      | 0.0592        | 0.0049     |
| deep_gbm               | 0.6801                   | 0.0713             | 0.1024         | 0.0117      | 0.0642        | 0.0047     |
| gradient_boosting      | 0.6748                   | 0.1194             | 0.1025         | 0.0184      | 0.0637        | 0.0078     |
| hist_gradient_boosting | 0.6685                   | 0.0477             | 0.1046         | 0.0085      | 0.0669        | 0.0041     |
| mlp                    | 0.668                    | 0.0845             | 0.1042         | 0.0159      | 0.0613        | 0.0072     |
| random_forest          | 0.6647                   | 0.0825             | 0.1047         | 0.013       | 0.0659        | 0.0053     |
| lightgbm               | 0.6626                   | 0.0523             | 0.1055         | 0.01        | 0.0672        | 0.0065     |
| ridge                  | 0.659                    | 0.0886             | 0.1056         | 0.015       | 0.0696        | 0.0071     |
| linear                 | 0.6582                   | 0.0891             | 0.1057         | 0.0151      | 0.0698        | 0.0072     |
| xgboost                | 0.6547                   | 0.1336             | 0.1054         | 0.0197      | 0.0657        | 0.0085     |
| saint                  | 0.6358                   | 0.073              | 0.1093         | 0.0114      | 0.0719        | 0.0063     |
| tab_cnn                | 0.62                     | 0.0953             | 0.1114         | 0.0149      | 0.078         | 0.0105     |
| ft_transformer         | 0.6064                   | 0.1076             | 0.1134         | 0.0181      | 0.0689        | 0.0111     |
| tabnet                 | 0.606                    | 0.0659             | 0.114          | 0.0112      | 0.0725        | 0.0051     |
| cart                   | 0.3653                   | 0.3557             | 0.141          | 0.0382      | 0.0891        | 0.0174     |
| elasticnet             | -0.0106                  | 0.0098             | 0.183          | 0.0067      | 0.1434        | 0.0081     |

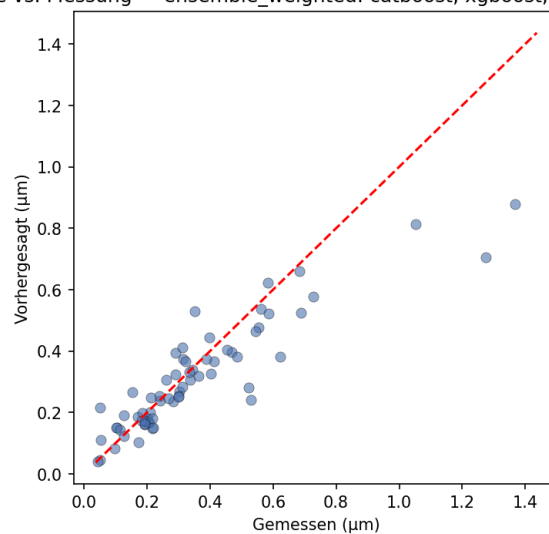
### 2 • Ergebnisse nach Hyperparameter-Optimierung (HPO)

Bester RMSE je Modell aus der Hyperparameter-Optimierung (Optuna, 5-fach CV). Das beste Einzelmodell ist **ensemble\_weighted: catboost, xgboost, tab\_cnn**.

| Modell                 | Bester RMSE (HPO)    |
|------------------------|----------------------|
| catboost               | 0.0909 $\mu\text{m}$ |
| xgboost                | 0.0949 $\mu\text{m}$ |
| tab_cnn                | 0.0957 $\mu\text{m}$ |
| hist_gradient_boosting | 0.0967 $\mu\text{m}$ |
| mlp                    | 0.0970 $\mu\text{m}$ |
| gradient_boosting      | 0.0970 $\mu\text{m}$ |
| extra_trees            | 0.0978 $\mu\text{m}$ |
| ft_transformer         | 0.0985 $\mu\text{m}$ |
| deep_gbm               | 0.1000 $\mu\text{m}$ |
| svr                    | 0.1002 $\mu\text{m}$ |
| lightgbm               | 0.1014 $\mu\text{m}$ |
| saint                  | 0.1027 $\mu\text{m}$ |
| random_forest          | 0.1039 $\mu\text{m}$ |
| ridge                  | 0.1055 $\mu\text{m}$ |
| elasticnet             | 0.1055 $\mu\text{m}$ |
| cart                   | 0.1153 $\mu\text{m}$ |
| tabnet                 | 0.1327 $\mu\text{m}$ |

Testset-Metriken **ensemble\_weighted: catboost, xgboost, tab\_cnn** (nach HPO, retrained):  
 $R^2 = 0.7548$  | RMSE = 0.1262  $\mu\text{m}$  | MAE = 0.0755  $\mu\text{m}$

Vorhersage vs. Messung — ensemble\_weighted: catboost, xgboost, tab\_cnn  $R^2=0.755$



Bestes Einzelmodell nach HPO: ensemble\_weighted: catboost, xgboost, tab\_cnn

### 3 · Ensembling — Alle Kombinationen

Die drei stärksten Einzelmodelle werden zu Averaging-, Weighted- und Stacking-Ensembles kombiniert. Die grün markierte Zeile ist das beste Ensemble.

| Ensemble  | Rmse   | Std    |
|-----------|--------|--------|
| weighted  | 0.0875 | 0.0082 |
| averaging | 0.0876 | 0.0082 |
| stacking  | 0.0898 | 0.0076 |

### 4 · Finaler Prädiktor

Der finale Prädiktor ist **ensemble:weighted** (gewählt nach Vergleich aller Einzelmodelle und Ensembles auf dem Testset).

| Kennzahl       | Wert              |
|----------------|-------------------|
| R <sup>2</sup> | <b>0.7548</b>     |
| RMSE           | <b>0.1262 µm</b>  |
| MAE            | <b>0.0755 µm</b>  |
| Modell         | ensemble:weighted |

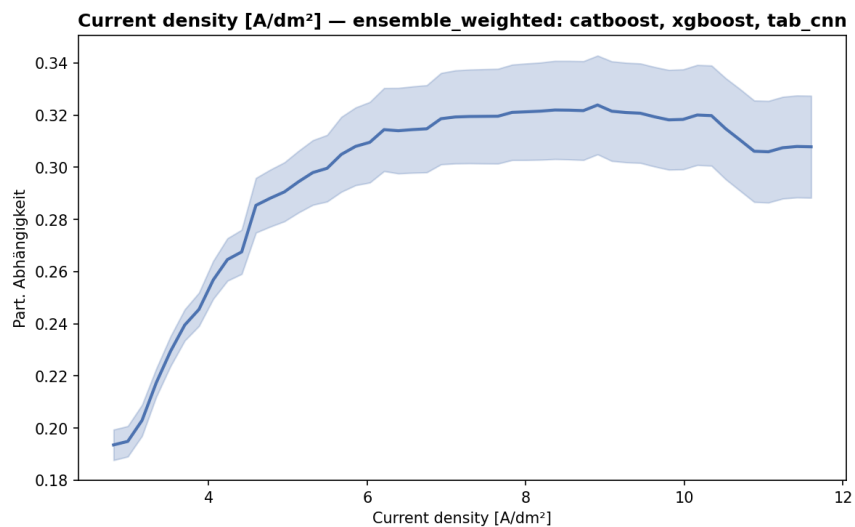
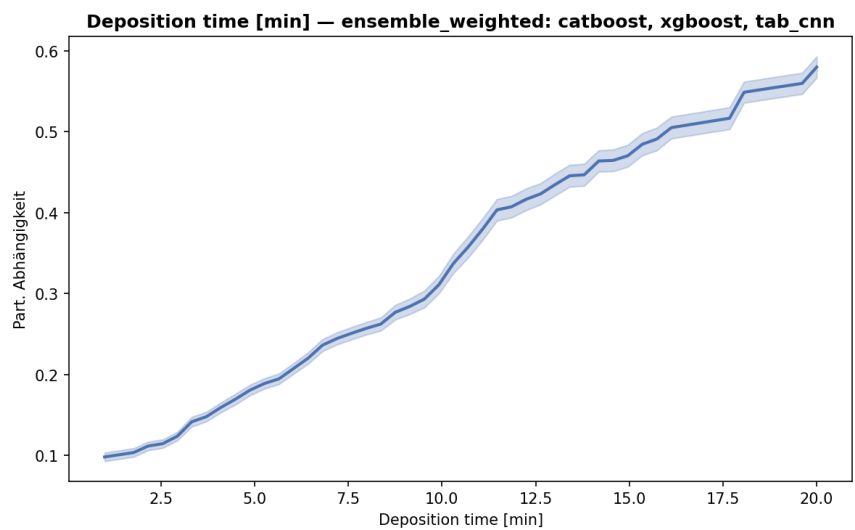
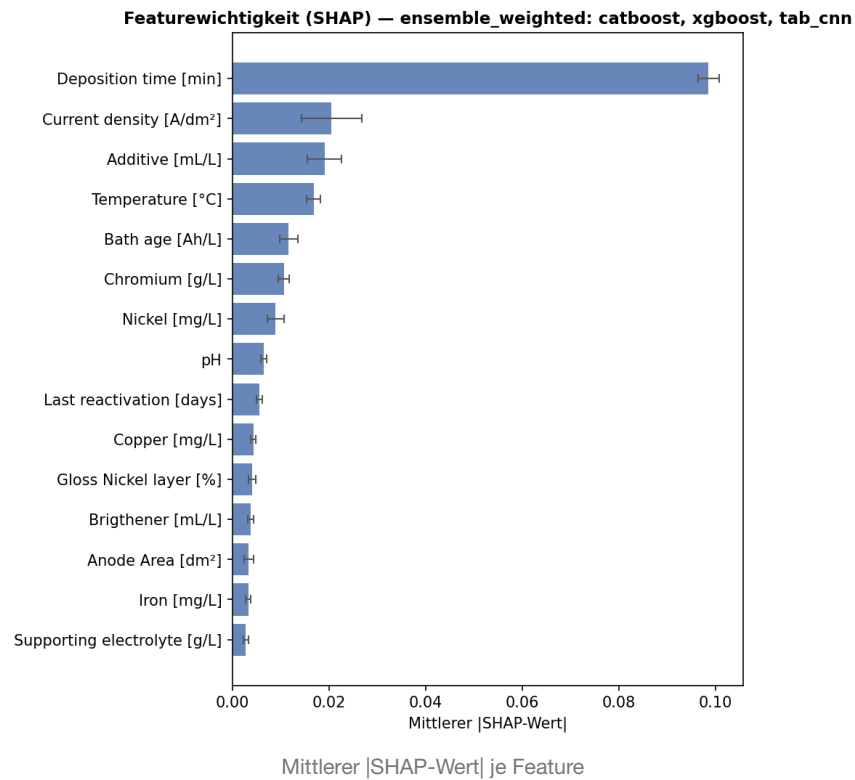
### 5 · Featurewichtigkeit (SHAP) & Partielle Abhängigkeitsplots (PDP)

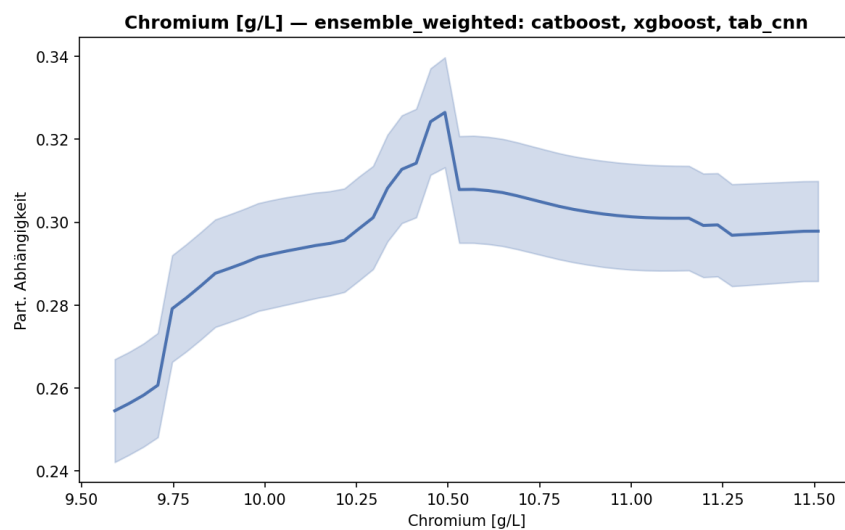
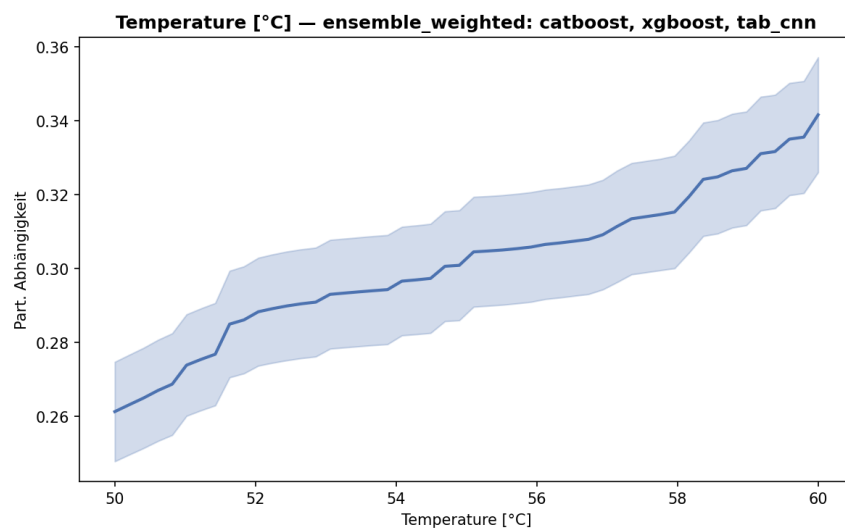
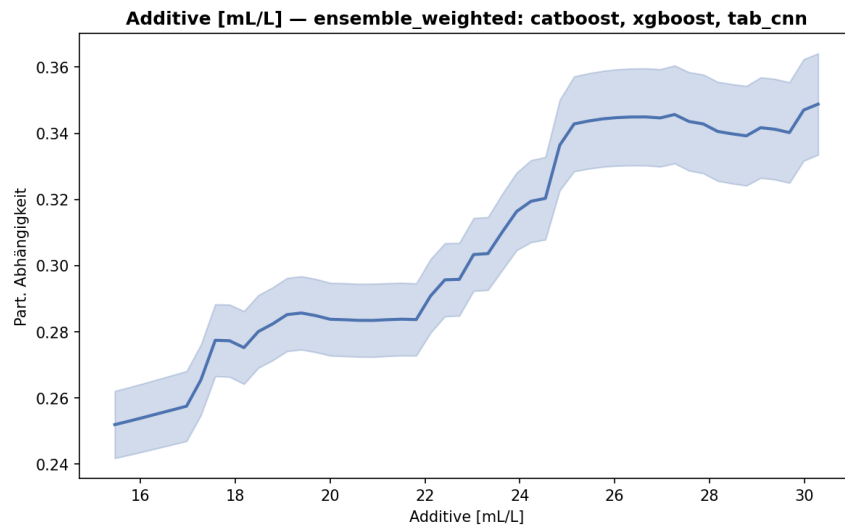
SHAP (SHapley Additive exPlanations) zeigt den Einfluss jedes Prozessparameters auf die Vorhersage. Fehlerbalken zeigen die Stabilität über Cross-Validation-Folds. PDP zeigt den durchschnittlichen Effekt bei Variation eines Parameters.

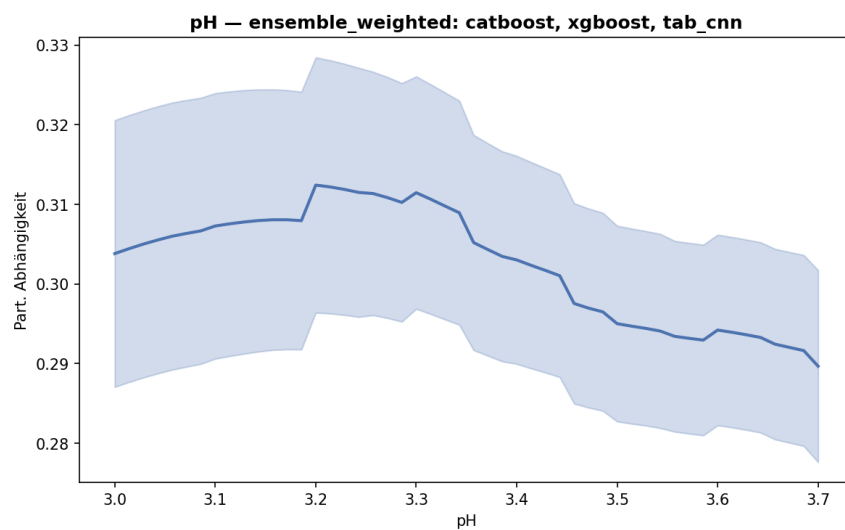
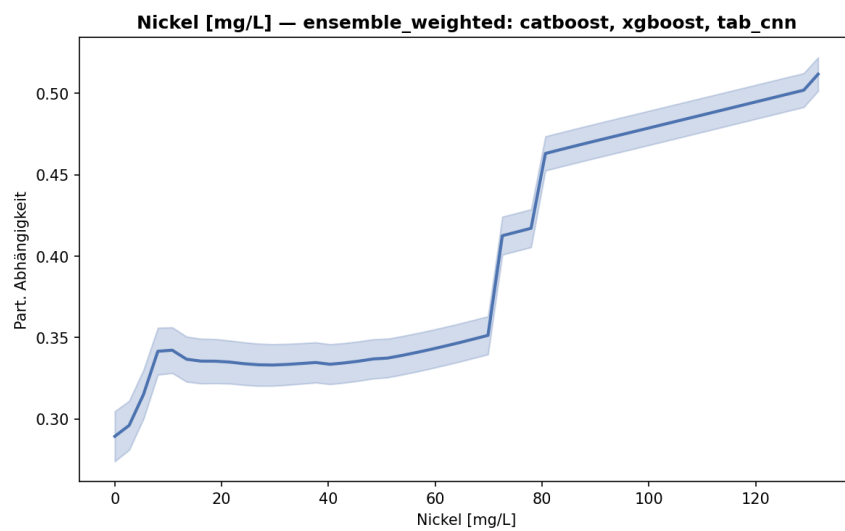
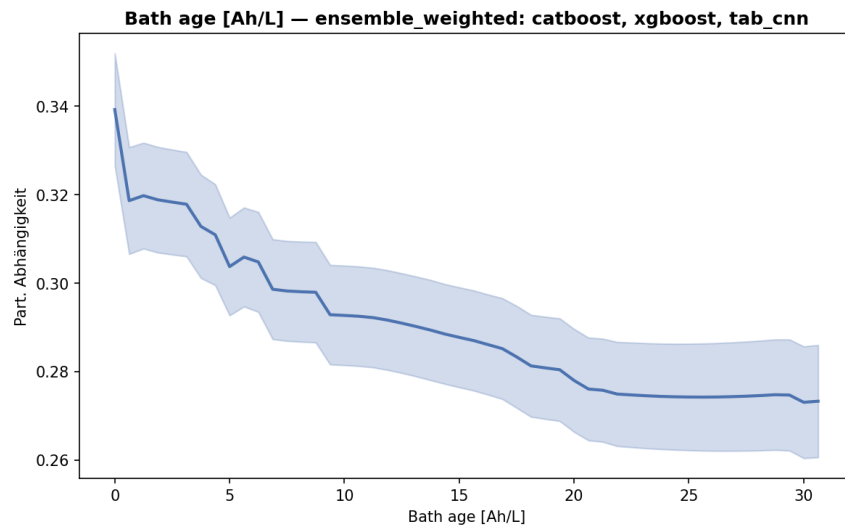
SHAP auf Testset: **ensemble:weighted**. Faltbasierte SHAP (±Std.-Abw.):

**ensemble\_weighted: catboost, xgboost, tab\_cnn** (bestes Einzelmodell). PDP:

**ensemble\_weighted: catboost, xgboost, tab\_cnn** über Cross-Validation-Folds.







## 6 · Inverse ML — Parametervorschläge

Bayesianische Suche nach Prozessparametern, die eine Zieldicke von **0.230  $\mu\text{m}$**  erzeugen. Modell: **ensemble:weighted**.

| Deposition time [min] | Current density [A/dm <sup>2</sup> ] | Chromium [g/L] | Buffer [g/L] | predicted_thickness | error  |
|-----------------------|--------------------------------------|----------------|--------------|---------------------|--------|
| 10.4979               | 10.6971                              | 10.0589        | 76.6995      | 0.2296              | 0.0004 |
| 9.3662                | 6.7389                               | 9.7846         | 75.7328      | 0.2286              | 0.0014 |
| 12.2559               | 3.2087                               | 10.7565        | 73.6759      | 0.2274              | 0.0026 |
| 9.6009                | 5.7922                               | 10.0587        | 79.867       | 0.2264              | 0.0036 |
| 9.333                 | 6.2288                               | 10.122         | 82.8172      | 0.2232              | 0.0068 |
| 11.375                | 3.0562                               | 10.4804        | 77.3617      | 0.2231              | 0.0069 |
| 9.6863                | 7.1341                               | 10.0825        | 83.45        | 0.2375              | 0.0075 |
| 9.5418                | 6.0477                               | 10.1315        | 82.082       | 0.2225              | 0.0075 |
| 9.4176                | 7.2733                               | 10.1404        | 84.5921      | 0.2203              | 0.0097 |
| 9.293                 | 7.2736                               | 10.0536        | 86.0832      | 0.2201              | 0.0099 |



## Klassifikation — Dünn / Dick (Schwellenwert: 0.2654 $\mu\text{m}$ )

Der binäre Klassifikator entscheidet, ob eine Schicht 'dünn' oder 'dick' ist.  
Schwellenwert = Median der Trainingsschichtdicken. Die Pipeline durchläuft dieselbe HPO- und Ensembling-Prozedur wie die Regression.

### 1 · Cross-Validation — Test aller Basisklassifikatoren

| Modell                 | Accuracy Mittel | Accuracy Std | F1 Mittel | F1 Std | AUC Mittel | AUC Std |
|------------------------|-----------------|--------------|-----------|--------|------------|---------|
| mlp                    | 0.8345          | 0.059        | 0.8372    | 0.0585 | 0.89       | 0.0512  |
| gradient_boosting      | 0.8371          | 0.0399       | 0.836     | 0.0445 | 0.917      | 0.0313  |
| logistic               | 0.8344          | 0.0482       | 0.8326    | 0.0467 | 0.9029     | 0.0418  |
| hist_gradient_boosting | 0.8317          | 0.0566       | 0.8325    | 0.0546 | 0.8923     | 0.0458  |
| deep_gbm               | 0.8237          | 0.0549       | 0.8231    | 0.0523 | 0.8896     | 0.0449  |
| catboost               | 0.821           | 0.0596       | 0.8201    | 0.0649 | 0.9078     | 0.0382  |
| random_forest          | 0.821           | 0.0525       | 0.8172    | 0.059  | 0.8904     | 0.0295  |
| xgboost                | 0.8156          | 0.0431       | 0.8156    | 0.0426 | 0.893      | 0.0291  |
| lightgbm               | 0.8103          | 0.0556       | 0.8086    | 0.0674 | 0.8952     | 0.0467  |
| tab_cnn                | 0.8102          | 0.0647       | 0.8079    | 0.0545 | 0.8709     | 0.0476  |
| c50                    | 0.8155          | 0.0322       | 0.8066    | 0.0502 | 0.8141     | 0.0323  |
| saint                  | 0.8049          | 0.0517       | 0.8045    | 0.0627 | 0.868      | 0.0513  |
| extra_trees            | 0.8023          | 0.0672       | 0.8003    | 0.0788 | 0.8692     | 0.0416  |
| cart                   | 0.8049          | 0.0583       | 0.7973    | 0.0683 | 0.8043     | 0.059   |
| tabnet                 | 0.7836          | 0.0631       | 0.7936    | 0.0638 | 0.876      | 0.0375  |
| ft_transformer         | 0.7676          | 0.0815       | 0.7623    | 0.0834 | 0.8579     | 0.0587  |

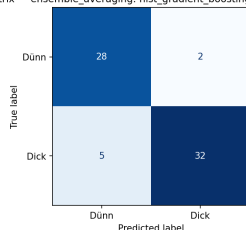
### 2 · Ergebnisse nach Hyperparameter-Optimierung (HPO)

Bester F1-Score je Klassifikator aus der HPO. Bestes Einzelmodell:  
**ensemble\_averaging: hist\_gradient\_boosting, ft\_transformer, tab\_cnn.**

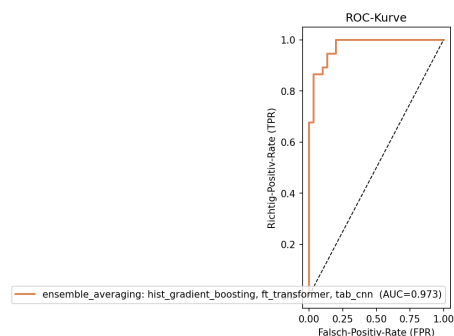
| Modell                 | Bester F1 (HPO) |
|------------------------|-----------------|
| hist_gradient_boosting | 0.8662          |
| ft_transformer         | 0.8595          |
| tab_cnn                | 0.8585          |
| deep_gbm               | 0.8563          |
| mlp                    | 0.8562          |
| gradient_boosting      | 0.8534          |
| cart                   | 0.8447          |
| logistic               | 0.8417          |
| random_forest          | 0.8412          |
| c50                    | 0.8390          |
| extra_trees            | 0.8357          |
| saint                  | 0.8275          |
| tabnet                 | 0.7278          |

Testset-Metriken **ensemble\_averaging: hist\_gradient\_boosting, ft\_transformer, tab\_cnn**  
(nach HPO, retrained): F1 = **0.9014** | AUC = **0.9730** | Accuracy = **0.8955**

Konfusionsmatrix — ensemble\_averaging: hist\_gradient\_boosting, ft\_transformer, tab\_cnn



Konfusionsmatrix — ensemble\_averaging:  
hist\_gradient\_boosting, ft\_transformer, tab\_cnn



ROC-Kurve — ensemble\_averaging:  
hist\_gradient\_boosting, ft\_transformer, tab\_cnn

### 3 · Ensembling — Alle Kombinationen

Averaging-, Weighted- und Stacking-Ensembles der drei stärksten Klassifikatoren.  
Grün markiert: bestes Ensemble (nach F1).

| Ensemble  | F1     | Std    |
|-----------|--------|--------|
| averaging | 0.8707 | 0.0282 |

| Ensemble | F1     | Std    |
|----------|--------|--------|
| weighted | 0.8707 | 0.0282 |
| stacking | 0.8672 | 0.0294 |

#### 4 · Finaler Klassifikator

Der finale Klassifikator ist **ensemble:averaging**.

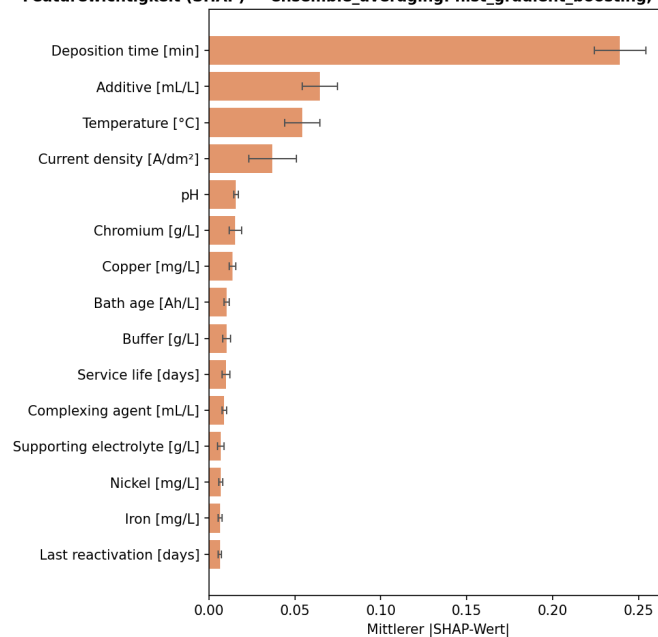
| Kennzahl | Wert               |
|----------|--------------------|
| F1-Score | <b>0.9014</b>      |
| AUC      | <b>0.9730</b>      |
| Accuracy | <b>0.8955</b>      |
| Modell   | ensemble:averaging |

#### 5 · Featureswichtigkeit (SHAP) & PDP

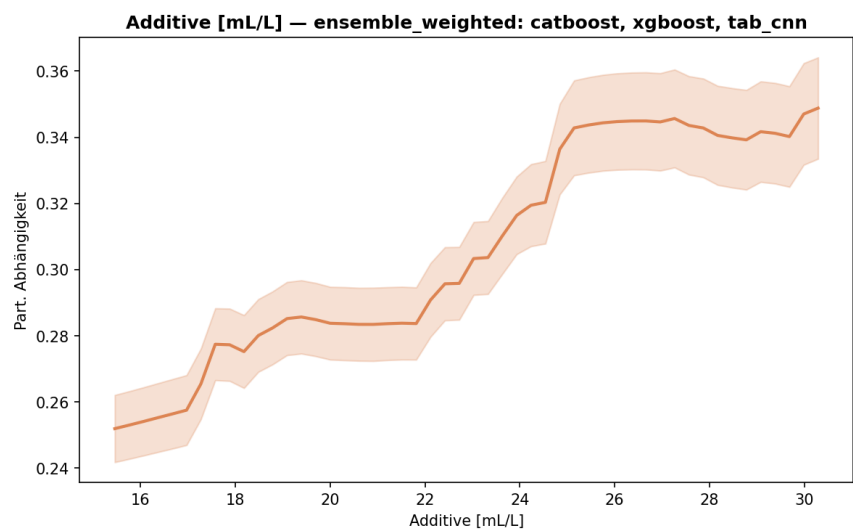
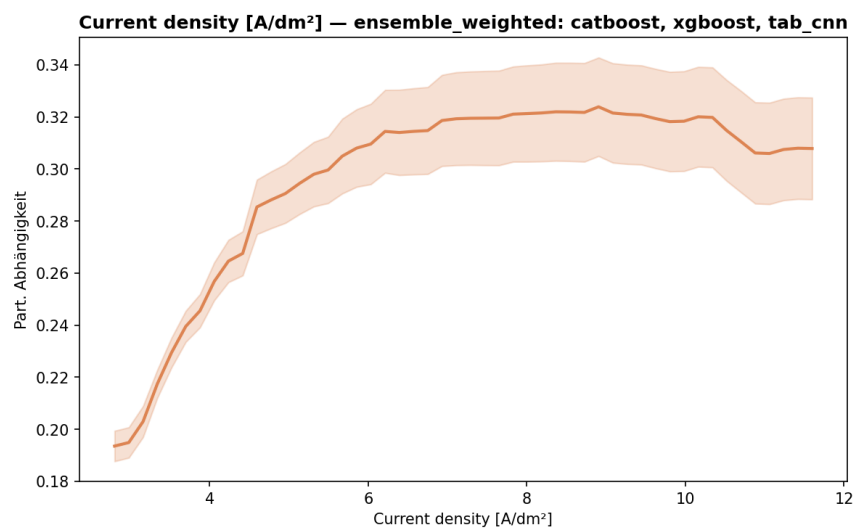
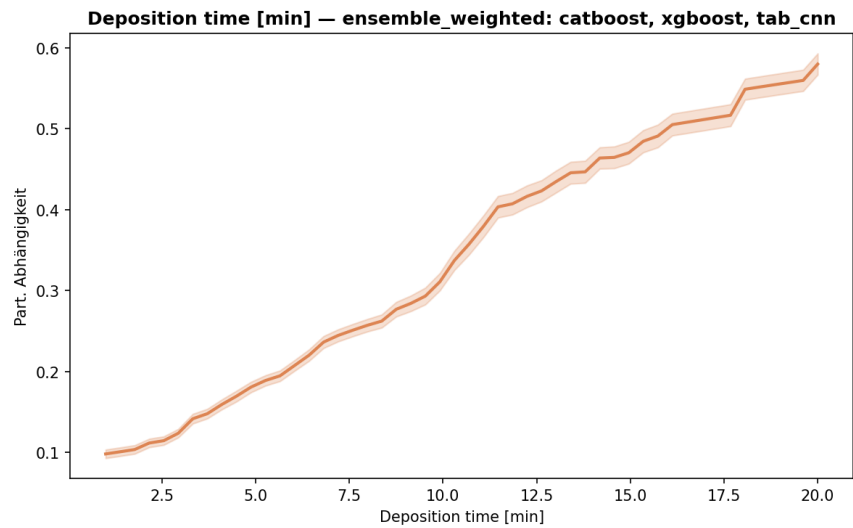
SHAP zeigt, welche Parameter die Dünn/Dick-Entscheidung am stärksten beeinflussen. PDP-Werte nahe 1,0 bedeuten: Modell erwartet Dickschicht; nahe 0,0: Dünnschicht.

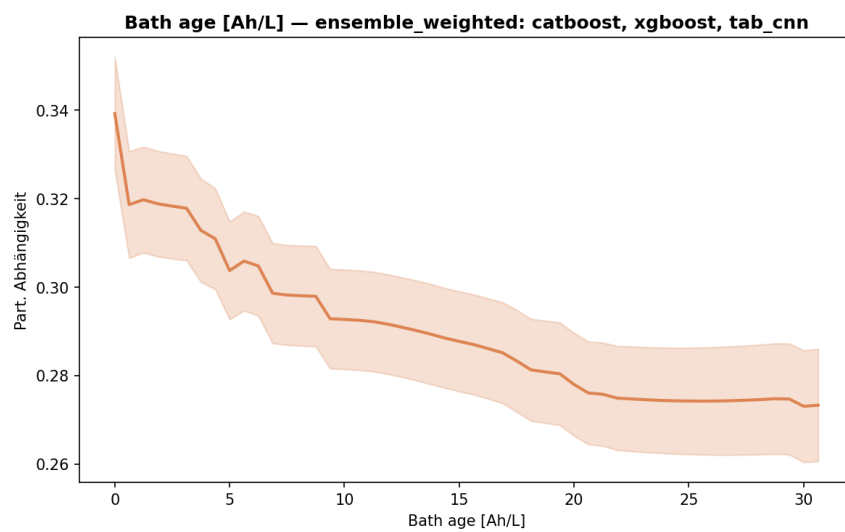
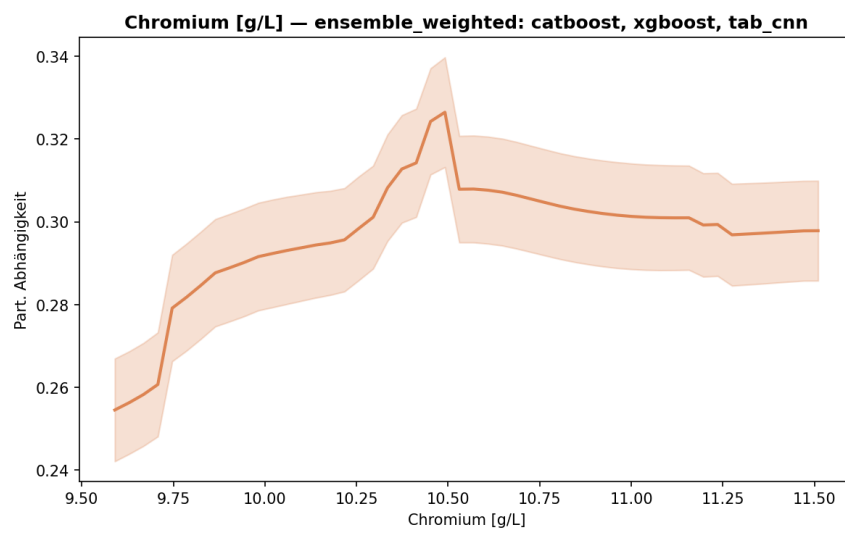
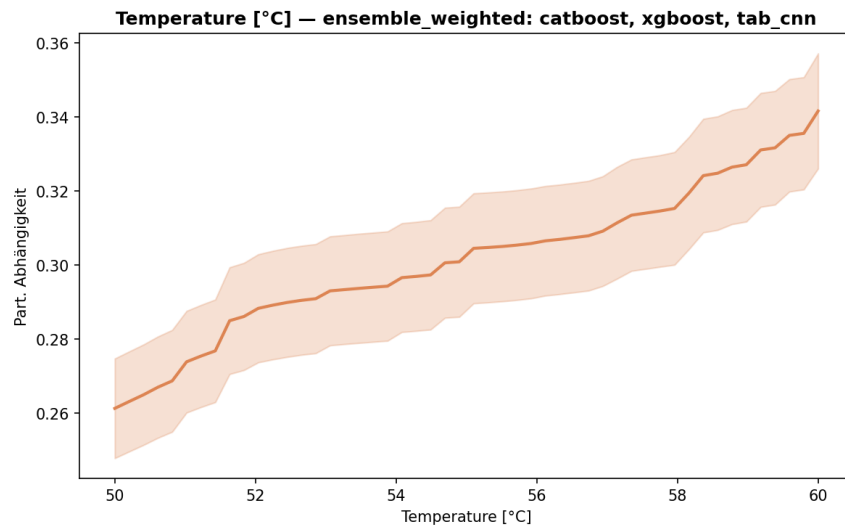
SHAP auf Testset: **ensemble:averaging**. Faltbasierte SHAP: **ensemble\_averaging: hist\_gradient\_boosting, ft\_transformer, tab\_cnn** (bestes Einzelmodell).

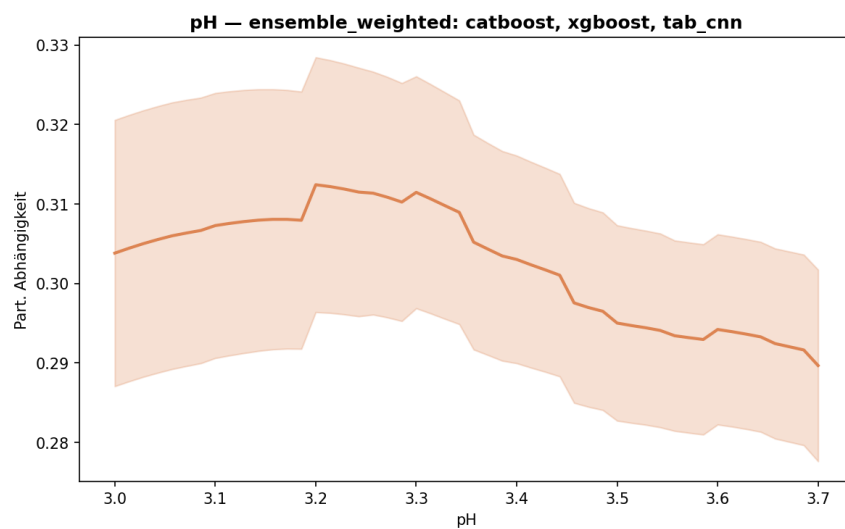
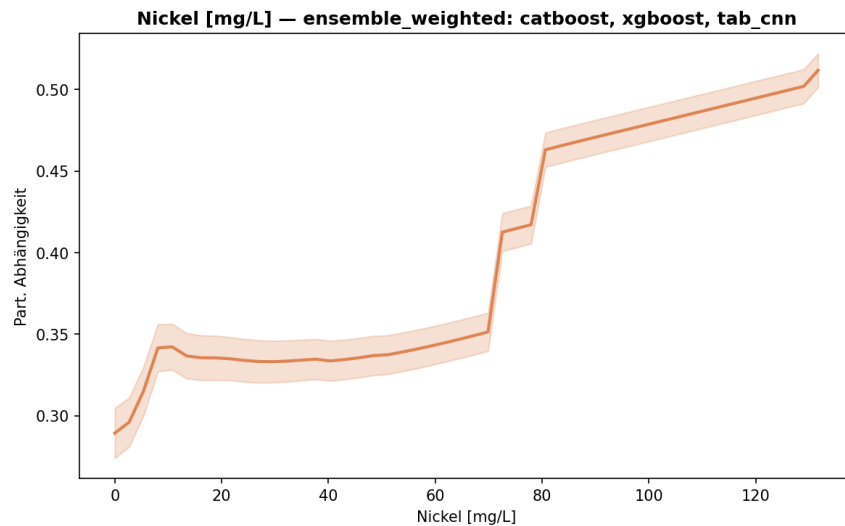
Featurewichtigkeit (SHAP) — ensemble\_averaging: hist\_gradient\_boosting, ft\_transformer, tab\_cnn



Mittlerer |SHAP-Wert| je Feature — Klassifikation







## 6 · Inverse ML — Parametervorschläge

Inverse ML läuft auf dem Regressionsmodell — nicht auf dem Klassifikator. Die Parametervorschläge basieren auf dem finalen Regressorprädiktor.

Zieldicke: **0.230  $\mu\text{m}$**  | Modell: **ensemble:weighted**

| Deposition time [min] | Current density [A/dm <sup>2</sup> ] | Chromium [g/L] | Buffer [g/L] | predicted_thickness | error  |
|-----------------------|--------------------------------------|----------------|--------------|---------------------|--------|
| 10.4979               | 10.6971                              | 10.0589        | 76.6995      | 0.2296              | 0.0004 |
| 9.3662                | 6.7389                               | 9.7846         | 75.7328      | 0.2286              | 0.0014 |
| 12.2559               | 3.2087                               | 10.7565        | 73.6759      | 0.2274              | 0.0026 |
| 9.6009                | 5.7922                               | 10.0587        | 79.867       | 0.2264              | 0.0036 |
| 9.333                 | 6.2288                               | 10.122         | 82.8172      | 0.2232              | 0.0068 |

| Deposition<br>time [min] | Current<br>density<br>[A/dm <sup>2</sup> ] | Chromium<br>[g/L] | Buffer<br>[g/L] | predicted_thickness | error  |
|--------------------------|--------------------------------------------|-------------------|-----------------|---------------------|--------|
| 11.375                   | 3.0562                                     | 10.4804           | 77.3617         | 0.2231              | 0.0069 |
| 9.6863                   | 7.1341                                     | 10.0825           | 83.45           | 0.2375              | 0.0075 |
| 9.5418                   | 6.0477                                     | 10.1315           | 82.082          | 0.2225              | 0.0075 |
| 9.4176                   | 7.2733                                     | 10.1404           | 84.5921         | 0.2203              | 0.0097 |
| 9.293                    | 7.2736                                     | 10.0536           | 86.0832         | 0.2201              | 0.0099 |